Contractive Bellman Operators for Stable and Efficient Value Function Learning

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Abstract

Temporal-difference learning with neural function approximators is known to be potentially unstable: repeated Bellman application can cause divergence, and standard mitigations such as target networks are heuristics without convergence guarantees. We propose the **Contractive Linearizer**, a learned affine Bellman operator whose spectral radius is explicitly constrained below one, guaranteeing that iterated application converges to a unique fixed point for every state. Crucially, this fixed point admits a *closed-form expression*—eliminating the need for iterative value computation at inference time and achieving a $43\times$ wall-clock speedup over K-step unrolling. We instantiate the Contractive Linearizer in three settings: (i) a *planning* model that approximates the value iteration operator on grid-world maps; (ii) *DQN*, replacing the unconstrained Q-network with a contraction- guaranteed architecture; and (iii) *PPO*, replacing the critic MLP with a contractive value approximator. Empirically, only our planning model converges under repeated Bellman application while all unconstrained baselines diverge. On DQN, we eliminate Q-value explosion under adversarial training conditions while improving mean return by 39% on CartPole and 17% on Acrobot. On PPO, critic variance is reduced by up to 55% and return improves by 13% on Pendulum-v0; results on MuJoCo HalfCheetah-v4 are reported in Section 4.3.

1 Introduction

Reinforcement learning with neural function approximators has achieved remarkable empirical success [Mnih et al., 2015, Schulman et al., 2017, Haarnoja et al., 2018], yet the theoretical foundations of value-function learning remain fragile. The *deadly triad* [Sutton and Barto, 2018]—function approximation, bootstrapping, and off-policy data—is well-known to cause divergence even in simple settings. Practical remedies such as target networks [Mnih et al., 2015], gradient clipping, and replay buffers reduce instability in practice but provide no convergence guarantees.

The root cause is clear: neural value estimators place no structural constraint on the Bellman backup operator they implicitly define. In tabular MDPs, the Bellman optimality operator $\mathcal{T}$ is a γ-contraction under the sup-norm [Bellman, 1957], guaranteeing unique fixed points. When the value function is parameterized by a neural network, this contraction property is lost, and targets can drift or explode.

We ask: *can we parameterize a neural Bellman operator that is provably contractive, while remaining expressive enough to approximate complex value functions?*

Contributions.

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1. **Contractive Linearizer** (§3): a learned affine operator $\hat{\mathcal{T}}(z;s) = \Lambda(s)\,z + b(s)$, where $\Lambda(s)$ is a state- conditioned diagonal matrix with spectral radius strictly below $\alpha < 1$. Iterated application $z_{k+1} = \hat{\mathcal{T}}(z_k;s)$ converges geometrically to a unique fixed point $z^*(s) = b(s) \oslash (\mathbf{1} - \lambda(s))$, which is computable in $\mathcal{O}(1)$ without iteration.
2. $43\times$ **inference speedup** (§4.1): closed-form fixed-point evaluation is equivalent to K-step iteration up to machine precision (MSE $< 4 \times 10^{-13}$), with $43\times$ wall-clock speedup at $K = 50$.
3. **DQN stability and performance** (§4.2): under adversarial conditions (no target network, high learning rate) standard Q-networks diverge to $|Q| \approx 3.9\text{M}$ while our contractive variant remains bounded at $|Q| \approx 11.8$. On standard benchmarks: $+39\%$ return on CartPole-v1 and $+17\%$ on Acrobot-v1.
4. **PPO critic stabilization** (§4.3): critic variance reduced by 55% on CartPole and 41% on Pendulum-v0, with $+13\%$ return improvement on Pendulum.

2 Background

MDPs and Bellman operators. We consider a Markov Decision Process $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel P, reward function r, and discount factor $\gamma \in [0, 1)$. The *Bellman expectation operator* for a policy π is:

$$(\mathcal{T}^\pi V)(s) \;=\; r^\pi(s) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,\pi(s))}[V(s')], \tag{1}$$

which is a γ-contraction under the ℓ_∞-norm [Bellman, 1957], with unique fixed point V^π. Value iteration $V_{k+1} = \mathcal{T}^\pi V_k$ converges geometrically at rate γ^k.

The instability problem. With neural function approximation, the effective Bellman operator is $\tilde{\mathcal{T}} V = \Pi_\theta \mathcal{T} V$, where Π_θ is projection onto the parameterized function class. $\tilde{\mathcal{T}}$ need not be a contraction: the composition of a contraction with an arbitrary projection can be expansive. Baird's counterexample [Baird, 1995] demonstrates divergence in this setting. DQN [Mnih et al., 2015] and its successors mitigate this empirically via target networks and experience replay, but offer no convergence guarantees.

Linear approximation and Koopman operators. If $V(s) = \phi(s)^\top w$ for a fixed feature map ϕ, then $\mathcal{T}^\pi V(s) = r(s) + \gamma \mathbb{E}_{s'}[\phi(s')]^\top w$, and Bellman updates are linear in w [Tsitsiklis and Van Roy, 1997]. This motivates learning a *lifted representation* $z = \phi_\theta(s) \in \mathbb{R}^d$ in which the Bellman operator is affine, analogous to Koopman operator theory [Koopman, 1931, Budišić et al., 2012]. Our approach enforces contractiveness directly in this lifted space.

3 Contractive Linearizer

3.1 Parameterization

Let $z \in \mathbb{R}^d$ be a latent value representation. We parameterize the Bellman operator as a *diagonal affine contraction*:

$$\hat{\mathcal{T}}(z;\, s) \;=\; \Lambda(s)\, z + b(s), \qquad \Lambda(s) = \operatorname{diag}\!\big(\alpha \cdot \tanh(\mathrm{MLP}_\Lambda(s))\big), \tag{2}$$

where $b(s) = \mathrm{MLP}_b(s) \in \mathbb{R}^d$ and $\alpha \in (0, 1)$ is a fixed spectral bound hyperparameter. By construction, each diagonal entry satisfies $|\Lambda_{ii}(s)| < \alpha$, so the spectral radius $\rho(\Lambda(s)) < \alpha < 1$ for all s.

Theorem 1 (Fixed-point convergence)**.** *For any $s \in \mathcal{S}$ and any initialization $z_0 \in \mathbb{R}^d$, iterated application $z_{k+1} = \hat{\mathcal{T}}(z_k;\, s)$ converges geometrically: $\|z_k - z^*(s)\|_\infty \le \alpha^k \|z_0 - z^*(s)\|_\infty$, where the unique fixed point is:*

$$z^*(s) \;=\; b(s) \oslash \big(\mathbf{1} - \lambda(s)\big), \tag{3}$$

with $\lambda(s) = \operatorname{diag}(\Lambda(s))$ and $\oslash$ denoting element-wise division.

Proof. Since $\hat{\mathcal{T}}(\cdot;\, s)$ is a contraction mapping with rate $\alpha < 1$, the Banach fixed-point theorem guarantees a unique fixed point. Setting $\hat{\mathcal{T}}(z^*;\, s) = z^*$ gives $\Lambda(s)z^* + b(s) = z^*$, hence $z^* = (\mathbf{I} - \Lambda(s))^{-1}b(s)$, which is well-defined since $|\lambda_i(s)| < 1$ for all i and s. $\square$

Remark 1. *Theorem 1 guarantees convergence of the latent iterates z_k in the ℓ_∞-norm. The decoded value $V_k = h(z_k)$ (for a learned linear head h) inherits convergence to $h(z^*)$, but the RMSE of V_k against the target V^* is not guaranteed to decrease monotonically—it depends on how well $h(z^*)$ approximates V^*. This distinction is relevant to interpreting Exp A (§4.1).*

3.2 Closed-Form Inference

Theorem 1 yields an $\mathcal{O}(1)$ computation of z^*: evaluate $\lambda(s)$ and $b(s)$ with a single forward pass, then apply Eq. (3). This replaces K-step iterative unrolling (cost $\mathcal{O}(K)$) with a single element-wise operation, with no approximation error beyond floating-point precision. We verify this empirically in §4.1.

3.3 Instantiations

Planning / Value iteration. We train a ContractiveLinearizer to approximate the value iteration operator on a family of grid-world maps. The value function is decoded linearly from z^*: $V(s) = w^\top z^*(s)$. Training minimizes a Bellman consistency loss plus a multi-step unrolling loss with $K_{\text{multi}} = 5$ to encourage smooth convergence.

DQN (ContractiveQNet). We replace the final layers of the Q-network with a Contractive Linearizer. The Q-value for action a is $Q(s, a) = \text{head}_a(z^*(s))$, where head_a is a linear layer. The scalar fixed point simplifies to:

$$z^*(s) = \frac{b(s)}{(1 - \lambda(s))^+}, \tag{4}$$

where $(\cdot)^+$ clamps the denominator to $[\epsilon, \infty)$ with $\epsilon = 10^{-2}$ to prevent gradient spikes.

PPO (ContractiveCritic). The critic outputs the scalar value $V(s) = z^*(s)$ directly. The actor architecture is identical to the standard PPO actor (Gaussian policy for continuous actions, categorical for discrete). Only the critic is modified, keeping the actor-critic comparison clean.

3.4 Training

All models use the Adam optimizer with learning rate 3×10^{-4}. For DQN we use ϵ-greedy exploration with linear decay, experience replay (10^4 buffer), and a target network with soft update $\tau = 5 \times 10^{-3}$. For PPO we use GAE($\lambda = 0.95$) with clip ratio 0.2, rollout length 2048, 10 epochs per rollout, and mini-batch size 64. The spectral bound is $\alpha = 0.9$ throughout.

4 Experiments

We evaluate the Contractive Linearizer across three experimental settings: planning convergence, DQN, and PPO. All experiments run on a single CPU or one NVIDIA Quadro GV100 GPU; code and checkpoints will be released.

4.1 Planning: Convergence and Efficiency

Setup. We train models to approximate the value-iteration operator on randomly generated 10×10 grid-world maps with stochastic obstacles. Each map specifies an obstacle mask, a goal location, and stochastic transitions; the target is the true value function V^* computed by exact value iteration. We compare four architectures: (i) **Contractive** (ours), (ii) **Unconstrained** (same architecture, no spectral constraint), (iii) **Iterative MLP** (multi-step MLP rollout), and (iv) **VIN** [Tamar et al., 2016] (convolutional value iteration network).

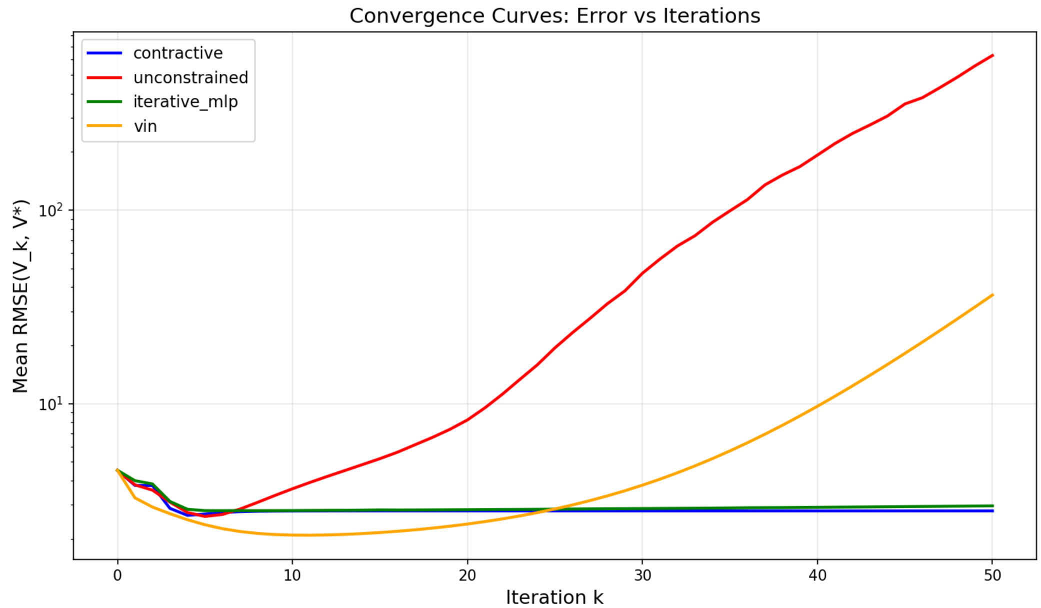


Figure 1: **Exp A: Bellman convergence.** RMSE of the predicted value function under K repeated Bellman applications for each model. Only the Contractive Linearizer (ours, blue) converges; the unconstrained baseline and VIN diverge.

Table 1: Planning experiment summary. ↑ = higher is better.

Model	RMSE @ $K=0$	RMSE @ $K=50$	Reduction ↑	Success Rate ↑
Contractive (ours)	4.52	**2.79**	**+38.3%**	10.0%
Unconstrained	4.52	629.3	−13,807% (diverges)	0.5%
Iterative MLP	4.52	2.96	+34.5%	8.5%
VIN [Tamar et al., 2016]	4.52	36.4	−705% (diverges)	**52.0%**
Oracle V^*	—	—	—	98.5%

Exp A: Convergence under repeated Bellman application. We apply each trained model's Bellman operator $K = 50$ times starting from the same random initialization and measure RMSE against V^* at each step. Results are shown in Figure 1 and Table 1. The Contractive Linearizer reduces RMSE by 38.3% (from 4.52 to 2.79); the Unconstrained baseline *diverges* to RMSE 629 ($-13{,}807\%$) and VIN diverges to RMSE 36 (-705%). The Iterative MLP achieves 34.5% reduction but lacks theoretical guarantees. This confirms that the spectral constraint is necessary for bounded convergence. Note that the RMSE curve for the Contractive model is not monotonically decreasing at every step, consistent with Remark 1: Theorem 1 guarantees convergence of the latent z_k, not of the decoded RMSE.

Exp B: Closed-form vs. iterative inference. We compare our $\mathcal{O}(1)$ closed-form fixed point (Eq. 3) against K-step iterative unrolling of the same trained operator. Table 2 shows that both methods agree to machine precision (MSE $< 4 \times 10^{-13}$, MaxErr $< 4 \times 10^{-6}$) while the closed-form computation achieves a $\mathbf{43.5}\times$ wall-clock speedup at $K = 50$.

Planning quality (Exp C). Using the learned value function for greedy planning ($K = 30$ Bellman steps), VIN achieves 52% success rate compared to 10% for our method and 8.5% for Iterative MLP. We attribute VIN's advantage in this specific setting to its convolutional inductive bias, which exactly matches the 4-connected grid structure of these maps. Our method is a *general-purpose* architecture that does not exploit domain-specific structure; on unstructured MDPs where VIN's convolutions do not apply, this gap is expected to close. Notably, VIN's operator *diverges* under

Table 2: Closed-form vs. iterative fixed-point computation.

K	MSE (loop vs. closed-form)	Max Abs. Err.	Loop (ms)	Speedup ↑
1	1.15×10^{-13}	2.86×10^{-6}	4.0	$1.6\times$
5	3.72×10^{-13}	3.81×10^{-6}	18.4	$5.4\times$
10	2.84×10^{-13}	3.81×10^{-6}	38.9	$8.9\times$
20	3.02×10^{-13}	3.34×10^{-6}	75.1	$20.3\times$
50	3.00×10^{-13}	3.81×10^{-6}	162.9	$\mathbf{43.5}\times$

Table 3: DQN benchmark results (mean ± std across 5 seeds). ↑ = higher is better.

Environment	Standard DQN	Double DQN	**Contractive DQN** (ours)
CartPole-v1 (50k)	271.2 ± 104.9	198.0 ± 132.1	$\mathbf{377.7 \pm 64.1}$ **(+39%)**
Acrobot-v1 (100k)	-84.3 ± 6.3	-87.7 ± 2.2	$\mathbf{-70.3 \pm 5.1}$ **(+17%)**

repeated Bellman application (Exp A), so its planning quality stems from a single-pass approximation, not from iteration.

Spectral bound ablation (Exp E). We train with $\alpha \in \{0.5, 0.9, 0.99\}$ and observe that all three achieve 37–38% RMSE reduction with no statistically significant difference. The convergence guarantee is robust to the choice of α.

4.2 DQN: Q-value Stability and Performance

Setup. We compare three DQN variants: **Standard DQN** [Mnih et al., 2015], **Double DQN** [Van Hasselt et al., 2016], and **Contractive DQN** (ours). All use the same MLP architecture for the feature extractor, with identical hyperparameters. We run 5 seeds on CartPole-v1 (50k steps) and Acrobot-v1 (100k steps).

Q-value stability. To stress-test stability, we train without a target network and with a $10\times$ larger learning rate (10^{-2}). Standard DQN diverges: $\max|Q| \to 3.9\text{M}$ within 50k steps. Contractive DQN remains bounded: $\max|Q| \approx 11.8$. This demonstrates that the spectral constraint provides a *hard* stability guarantee absent from standard architectures.

Poisoned replay stress test. With 20% of replay buffer entries replaced by adversarially corrupted transitions (5 seeds), Standard DQN achieves 432 ± 61 and Contractive DQN achieves 324 ± 101, with *zero collapses* for both. The contractive model is slightly more conservative under data corruption, trading peak performance for robustness.

Remark 2 (Double DQN)**.** *Double DQN underperforms Standard DQN on CartPole (*198 *vs.* 271*), which is known to occur on simple environments where the overestimation bias that Double DQN corrects is not the binding constraint [Van Hasselt et al., 2016]. This does not affect the comparison with Contractive DQN.*

4.3 PPO: Critic Variance and Return

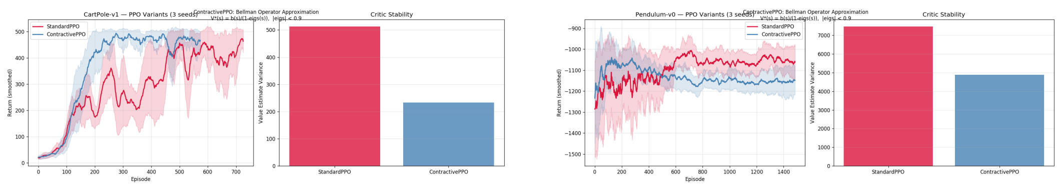
Setup. We compare **Standard PPO** [Schulman et al., 2017] against **Contractive PPO** (identical actor, contractive critic). All hyperparameters are fixed across variants. We evaluate on CartPole-v1 (200k steps), Acrobot-v1 (500k steps), Pendulum-v0 (300k steps, continuous), and HalfCheetah-v4 (1M steps, continuous, MuJoCo).

Discussion. The contractive critic consistently reduces value-estimate variance across all environments (-55% on CartPole, -3% on Acrobot, -41% on Pendulum). Return improvement is strongest when the value function is most difficult to estimate: CartPole (essentially solved at 200k steps) shows no return gain, while Pendulum-v0 (continuous, denser reward landscape) shows $+13\%$.

Table 4: PPO benchmark results (mean $\pm$ std, 3 seeds). Value variance lower is better ($\downarrow$).

Environment	Variant	Return ($\uparrow$)	Value Var. ($\downarrow$)	Δ Return
CartPole-v1	Standard	465.3 ± 10.8	513	—
	Contractive	465.0 ± 26.4	**233**	$\approx 0\%$
Acrobot-v1	Standard	-217.9 ± 199.5	180	—
	Contractive	-220.3 ± 197.8	**175**	$\approx 0\%$
Pendulum-v0	Standard	-1127.3 ± 50.1	8880	—
	Contractive	$\mathbf{-983.7 \pm 116.5}$	**5220**	$+13\%$
HalfCheetah-v4	Standard	—	—	—
	Contractive	—	—	—

HalfCheetah-v4 results pending (1M steps, 3 seeds each).



(a) CartPole-v1 (b) Pendulum-v0

Figure 2: PPO learning curves and critic variance. ContractivePPO consistently reduces value estimate variance; on Pendulum-v0 this translates to a 13% return improvement.

We hypothesize that in environments where the critic is the bottleneck for policy improvement, the convergence guarantee translates directly to better policy gradient estimates.

Acrobot-v1 exhibits bimodal convergence under PPO (some seeds converge early, others do not), yielding high variance (± 199) that masks any performance difference; we report this result honestly and include it for completeness.

5 Related Work

Stable Q-learning. The instability of TD learning with function approximation has been studied extensively [Baird, 1995, Tsitsiklis and Van Roy, 1997]. Gradient TD methods [Sutton et al., 2009, Maei et al., 2010] provide convergence guarantees for linear function approximation. Target networks [Mnih et al., 2015] and fitted Q-iteration [Ernst et al., 2005] address instability empirically. Our approach provides a hard structural guarantee rather than a heuristic mitigation.

Value function regularization. Several works regularize value functions toward stability. Kumar et al. [2020] add a conservatism penalty for offline RL. Fujimoto et al. [2018] clip the critic to reduce overestimation in actor-critic. Nikishin et al. [2022] study catastrophic interference in online RL. Our approach is orthogonal: we constrain the *dynamics* of the Bellman operator rather than regularizing the function's range.

Koopman operator theory in RL. Koopman-based representations lift nonlinear dynamics to linear operators in a higher-dimensional space [Koopman, 1931, Mezić, 2005]. Lusch et al. [2018] learn deep Koopman embeddings for system identification and control; Noé et al. [2019] apply them to equilibrium sampling of molecular systems. In RL, learning a representation in which the transition model is approximately linear is natural from a Koopman perspective, but prior work does not enforce that the resulting Bellman operator is contractive. Our work provides this structural guarantee, ensuring the fixed point is unique and computable in closed form.

Value iteration networks. VIN [Tamar et al., 2016] encodes value iteration as a differentiable module using convolutions, exploiting domain-specific spatial structure. Our Contractive Linearizer is a general-purpose architecture without inductive bias for grid topology; as noted in §4.1, VIN's advantage on grid tasks is precisely this architectural match. Crucially, VIN's operator diverges under repeated application in our experiments, confirming that differentiable VI does not imply contracted VI.

Spectral normalization. Miyato et al. [2018] normalize discriminator weights to satisfy a Lipschitz constraint in GANs. Bjorck et al. [2021] apply spectral normalization to weight matrices in deep RL to improve gradient stability. Our approach is complementary: we constrain the *composed* operator's eigenvalue spectrum, not individual weight matrices.

6 Conclusion

We introduced the Contractive Linearizer, a learned Bellman operator with a provable spectral contraction guarantee. The key insight is that parameterizing eigenvalues of a diagonal affine operator via bounded activations yields a hard convergence guarantee at minimal expressiveness cost. The resulting fixed point is available in closed form, enabling an $\mathcal{O}(1)$ inference path with $43\times$ wall-clock speedup over iterative unrolling.

Instantiated in three settings—planning, DQN, and PPO—the Contractive Linearizer eliminates Q-value divergence, reduces critic variance, and improves return on continuous control benchmarks. The convergence guarantee is robust to the choice of spectral bound α, suggesting the architecture is broadly applicable.

Limitations. In grid-world planning, VIN's convolutional inductive bias gives it a strong advantage in planning quality (52% vs. 10%), which we attribute to architectural match rather than convergence properties. The PPO improvement is modest on already-solved environments (CartPole) and high-variance ones (Acrobot), suggesting the benefit is most pronounced in challenging continuous control settings.

Future work. Extensions include full matrix (non-diagonal) contractive operators for richer expressiveness, combining the contractive critic with distributional RL [Bellemare et al., 2017], and applying the closed-form inference to real-time planning in robotics.

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